

Ideation Phase — Empathize & Discover

Date	07 March 2026
Team ID	
Project Name	Insurance Fraud Detection Using Machine Learning
Maximum Marks	4 Marks

Empathy Map Canvas

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviors and attitudes. It is a useful tool that helps teams better understand their users. Creating an effective solution requires understanding the true problem and the person who is experiencing it.

The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges. For this project, the primary user is the Insurance Claims Analyst / Fraud Reviewer who must validate insurance claims daily.

Primary User: Insurance Claims Analyst / Fraud Reviewer



Empathy Map Details

Dimension	Insights
THINK & FEEL (What really counts)	Deeply concerned about the rising number of fraudulent insurance claims causing financial losses. Worried about missing real fraud cases (false negatives). Wants a reliable, explainable tool they can confidently use for critical decision-making. Feels the weight of responsibility when approving large claim payouts.

Dimension	Insights
HEAR (What they hear)	Management pressuring teams to reduce claim losses by at least 30%. Colleagues discussing increasingly sophisticated fraud rings targeting auto and health insurance. Industry news reporting surge in post-disaster property fraud. Compliance team emphasizing audit requirements and documentation.
SEE (What they observe)	Thousands of insurance claims flooding in daily across automobile, health, and property segments. Complex, inconsistent claim forms with missing or contradictory data fields. Other analysts spending hours on manual cross-checks using spreadsheets. No standardized fraud scoring — outcomes vary by reviewer.
SAY & DO (Behaviour & actions)	Manually cross-checks claim documents against historical records. Escalates borderline or suspicious cases to senior reviewers. Relies heavily on personal experience and gut feeling rather than data. Spends 2–4 hours per complex claim. Often frustrated by lack of tooling.
PAIN (Fears & frustrations)	Slow, tedious manual review process. High false positive rate causing legitimate claims to be rejected. No data-driven decision support tool. Risk of missing sophisticated fraud patterns hidden across multiple claims. Fear of errors leading to regulatory non-compliance.
GAIN (Wants & needs)	Fast, automated fraud scoring with sub-2-second response time. Consistent, data-driven ML-based decisions with expandability. Significant reduction in the company's financial losses. Clear audit trail for every prediction. Ability to focus on high-value, genuinely ambiguous cases.

User Background & Context

Attribute	Details
Role	Insurance Claims Analyst / Fraud Reviewer
Domain	Automobile, Health, and Property Insurance
Experience Level	3–10 years in insurance claims processing
Technical Literacy	Moderate — comfortable with web tools, not a data scientist
Primary Pain Point	Manual, error-prone fraud detection consuming most of the workday
Primary Goal	Detect fraud faster and more accurately to protect company revenue
Device / Access	Desktop web browser (primary), occasionally mobile for alerts
Success Metric	Reduction in fraud payout losses; faster claim resolution time